

Probability & Statistics MCQs

Comprehensive Answer Key & Conceptual Guide

Part 1: Multiple Choice Questions

1. Probability Does Not Guarantee Outcomes

An engineer says: 'The probability that a sensor fails tomorrow is 20%. Therefore, if we monitor 10 identical sensors tomorrow, exactly 2 sensors will fail.' Is this reasoning correct?

Correct Answer: C — No, probability describes likelihood, not a guaranteed number of outcomes

2. Probability vs Certainty

A machine has a 90% probability of successfully completing a task. Which statement is correct?

Correct Answer: C — The machine is more likely to succeed than fail, but a particular attempt can still fail

3. Small Probability $\neq$ Impossible

An engineer says: 'The probability of a critical system failure is 0.001%. Therefore, the system cannot realistically fail.' What is wrong with this statement?

Correct Answer: C — A small probability means the event is unlikely, not impossible

4. Large Probability $\neq$ Guaranteed

An AI model predicts a 99% probability that a component will pass inspection. Which statement is most appropriate?

Correct Answer: B — The component has a high predicted likelihood of passing

5. The Law of Averages

An engineer observes that a machine has failed four times in a row. He says: 'The machine has failed four times, so it is now less likely to fail on the fifth attempt.' Assuming each failure is independent, what is wrong?

Correct Answer: B — The previous failures do not change the probability of the next failure

6. Coin Toss Misconception

A fair coin has produced: H H H H H. A student says: 'The next toss is more likely to be T because the coin needs to balance the results.' What is the correct response?

Correct Answer: C — Incorrect — for independent tosses, the next toss is still 50% Head and 50% Tail

7. Mean Does Not Describe Everything

An engineer says: 'The average temperature recorded by a sensor was 30°C. Therefore, the sensor normally operates around 30°C.' Can this conclusion necessarily be made?

Correct Answer: B — No, the mean alone does not describe the distribution or spread of the data

8. Same Mean, Different Systems

Two manufacturing systems have the following measurements:

System A: 49, 50, 50, 51, 50

System B: 10, 30, 50, 70, 90

Both have the same mean. Which conclusion is correct?

Correct Answer: B — System A has much less variability than System B

9. Mean vs Median

An engineer measures the salaries of employees in a company. One CEO earns ₹10 crore while most employees earn around ₹8 lakh. Which measure is likely to better represent the salary of a 'typical' employee?

Correct Answer: B — Median

10. Outlier Trap

A sensor records: 24.9, 25.1, 25.0, 25.2, 25.1, 80.0. An engineer calculates the mean and obtains a value considerably higher than 25°C. What is the most likely explanation?

Correct Answer: B — The extreme observation has strongly influenced the mean

11. Correlation ≠ Causation

An engineer observes: 'When server temperature increases, network errors also increase.' He concludes: 'Higher temperature causes network errors.' What is the problem?

Correct Answer: A — Correlation alone does not establish causation

12. Sample ≠ Population

An engineer tests 100 batteries from a production line and finds that their average lifetime is 8 hours. He concludes: 'Every battery produced by this factory has a lifetime of exactly 8 hours.' What is wrong?

Correct Answer: A — A sample average does not imply every individual has the same value

13. Random Sample

A company wants to estimate customer satisfaction. An engineer surveys only the 10 customers who voluntarily respond to an email and uses their responses to represent all customers. What is the main concern?

Correct Answer: A — The sample may be biased

14. Bigger Sample ≠ Automatically Correct

An engineer says: 'Our sample contains one million observations, so our conclusion must be correct.' Which response is best?

Correct Answer: B — Incorrect — a large but biased sample can still produce misleading conclusions

15. Standard Deviation

Two sensors have:

Sensor A: Mean = 50, SD = 1

Sensor B: Mean = 50, SD = 10

Which statement is most appropriate?

Correct Answer: B — Sensor B has greater variability

16. Standard Deviation $\neq$ Error

An engineer says: 'The standard deviation of the sensor readings is 5°C , so the sensor is wrong by 5°C .' What is wrong?

Correct Answer: A — Standard deviation measures the spread of observations; it is not automatically measurement error

17. Probability from Frequency

A factory observes that 50 out of 1,000 components failed during testing. An engineer says: 'The probability of failure is exactly 5%.' Which is the most careful interpretation?

Correct Answer: A — The observed failure proportion is 5%; it may be used as an estimate of the underlying failure probability under suitable assumptions

18. Past Data and Future Probability

An engineer analyzes 10 years of machine failures and estimates that the probability of failure next year is 3%. What assumption is implicitly important?

Correct Answer: A — The future operating conditions are sufficiently comparable to those represented by the historical data

19. Conditional Probability

An autonomous vehicle has a sensor that detects pedestrians. The engineer asks: 'What is the probability that the pedestrian detection system raises an alarm given that a pedestrian is actually present?' What type of probability is this?

Correct Answer: B — $P(\text{Alarm} \mid \text{Pedestrian})$

20. Reversing Conditional Probability

A medical AI has: $P(\text{Positive test} \mid \text{Disease}) = 95\%$. A student concludes: $P(\text{Disease} \mid \text{Positive test}) = 95\%$. Is this necessarily true?

Correct Answer: B — No, the two conditional probabilities are generally different

21. Rare Event Trap

A disease occurs in only 1 out of every 1,000 people. A diagnostic test is highly accurate. A student says: 'If the test is positive, the patient almost certainly has the disease.' What additional information is important before making this conclusion?

Correct Answer: A — Disease prevalence and the test's false-positive rate

22. Independence

Two machines are operating in the same factory. An engineer says: 'Their failures are independent because they are two different machines.' Is this necessarily correct?

Correct Answer: B — No, a common factor such as power failure or environmental conditions could make their failures dependent

23. Randomness

A sequence of measurements looks irregular: 12, 19, 14, 25, 17, 21, 13... An engineer says: 'The sequence looks random, so the measurements must be independent.' Is that conclusion justified?

Correct Answer: B — No, apparent randomness does not establish statistical independence

24. Expected Value

An engineering project has two possible outcomes:

Gain ■10,000 with probability 0.6

Lose ■5,000 with probability 0.4

An engineer says: 'Because there is a 40% chance of losing money, the project is a bad investment.' What is missing from this reasoning?

Correct Answer: A — The possible outcomes and their probabilities should be considered together

25. Expected Value $\neq$ Guaranteed Result

Suppose the expected lifetime of a particular component is 10 years. An engineer says: 'Therefore, every component should last exactly 10 years.' What is wrong?

Correct Answer: A — Expected value describes a long-run average, not the guaranteed lifetime of an individual component

26. Probability Distribution

An engineer says: 'A probability distribution tells us exactly what value will occur next.' Which is the correct response?

Correct Answer: B — No, it describes how probability is distributed across possible outcomes

27. Normal Distribution Trap

A dataset of manufacturing measurements approximately follows a bell-shaped curve. An engineer says: 'Because the distribution looks normal, every measurement must lie close to the mean.' What is wrong?

Correct Answer: A — A normal distribution still allows observations far from the mean

28. Statistics and Uncertainty

An engineer calculates: Mean = 50.2, Standard deviation = 0.3. He says: 'We now know the exact value of the population mean.' Is this correct?

Correct Answer: B — No, a sample statistic provides information about the population but generally involves uncertainty

29. Statistical Significance

An engineer observes that Method A produces slightly better results than Method B. He says: 'Method A is better because its average performance is higher.' What should he consider before making a strong conclusion?

Correct Answer: A — Whether the observed difference could reasonably be due to sampling variation

30. The Ultimate Conceptual Question

Based on 10,000 observations, the average failure rate was 2%.

Question 1: What was the observed failure rate in the collected data?

Question 2: What is the probability that the next randomly selected component will fail?

Correct Answer: C — Question 1 is statistics; Question 2 is probability

Part 2: Case Study — IoT Device Failure Analysis

Scenario: A company has 100,000 IoT devices deployed in the field. Historical records show that 3,000 devices failed during the last year.

- **What do we know from the data?**

We know the historical reality: exactly 3,000 specific devices from the deployed batch failed over the course of the last year.

- **What don't we know?**

We do not know the individual causes of failure, the exact timestamps of when they dropped off, or if the surviving 97,000 devices are close to wear-out thresholds.

- **What is a statistic here?**

The calculated sample failure rate, which is 3% ($3,000 / 100,000$ units).

- **What would be the population?**

The population is the theoretical infinite stream of all IoT devices manufactured under this specification operating under similar baseline conditions over time.

- **What would be the sample?**

The specific group of 100,000 deployed devices monitored over that exact one-year historical tracking frame.

- **Can we say the probability of failure is 3%?**

We can use 3% as an empirical estimate of the underlying probability, but it is not an absolute guaranteed property for all future conditions.

- **If we select one device tomorrow, what does '3% probability of failure' actually mean?**

It quantifies our uncertainty. It means that if we were to run this operational day thousands of times over, the device would be expected to fail in about 3% of those parallel runs.

- **Does 3% probability mean exactly 3 out of the next 100 devices will fail?**

No. Due to random binomial variations, exactly 3 might fail, or 0, 1, 5, or more could fail in any distinct batch of 100.

- **What assumptions would we need to make?**

We assume that device failures are statistically independent (one failing doesn't spark another to crash), and that environmental and manufacturing traits remain completely stable.

- **If the operating environment changes dramatically, can we still use the historical 3%?**

No. Probability models depend entirely on environment consistency. A hostile change breaks the model, making historical statistics unreflective of future risks.

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